SIES Graduate School of Technology
RISE WITH EDUCATION

EXPERIMENT NO: 01

Study of different network topologies using Cisco Packet Tracer

SIES Graduate School of Technology
RISE WITH EDUCATION

EXPERIMENT NO. 1

Title: Configuration and Comparison of Network Topologies Using Cisco Packet Tracer

Mapped Lab Outcome: LO3

Aim: Design, configure, and analyze different network topologies using simulation tools.

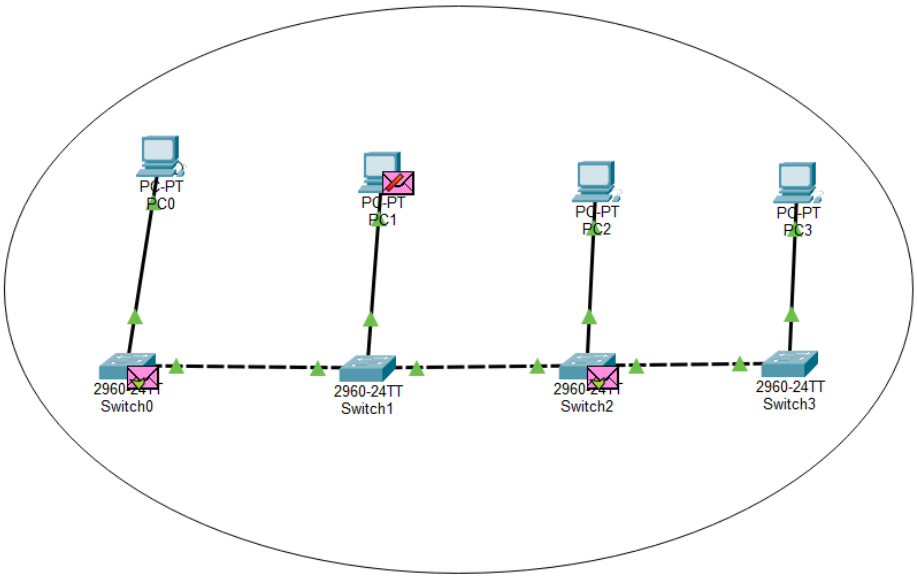
Objectives:

- Configure bus, star, ring, and mesh topologies.
- Simulate data transmission in different topologies.
- Compare performance, reliability, and scalability.
- Analyze fault tolerance of each topology.
- Understand practical applications of network topologies.

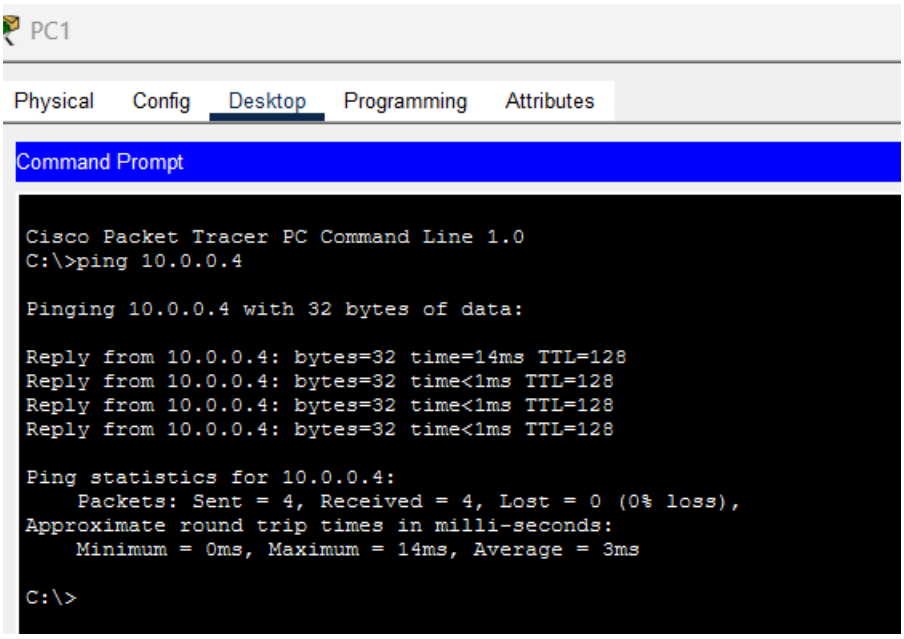
EQUIPMENT/SOFTWARE: Cisco Packet Tracer 7.3.0

1. **BUS TOPOLOGY:**

SCENARIO

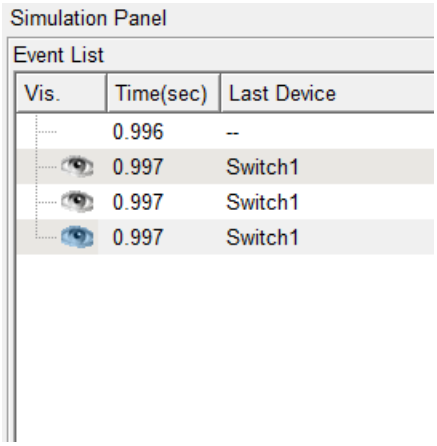


PING RESPONSE:



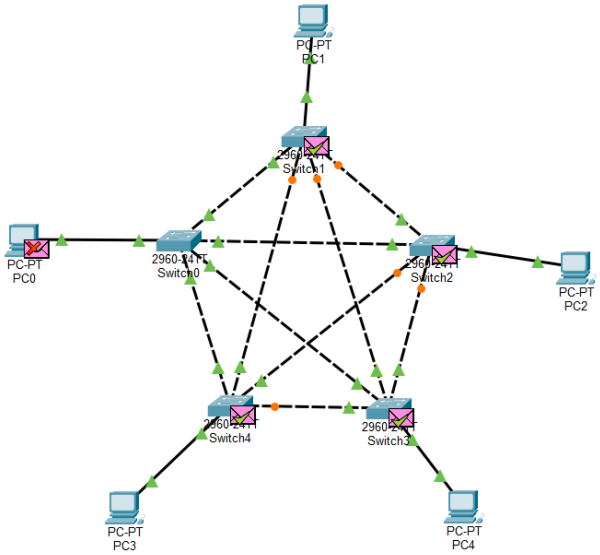
SIES Graduate School of Technology
RISE WITH EDUCATION

PACKET FLOW SEQUENCE:

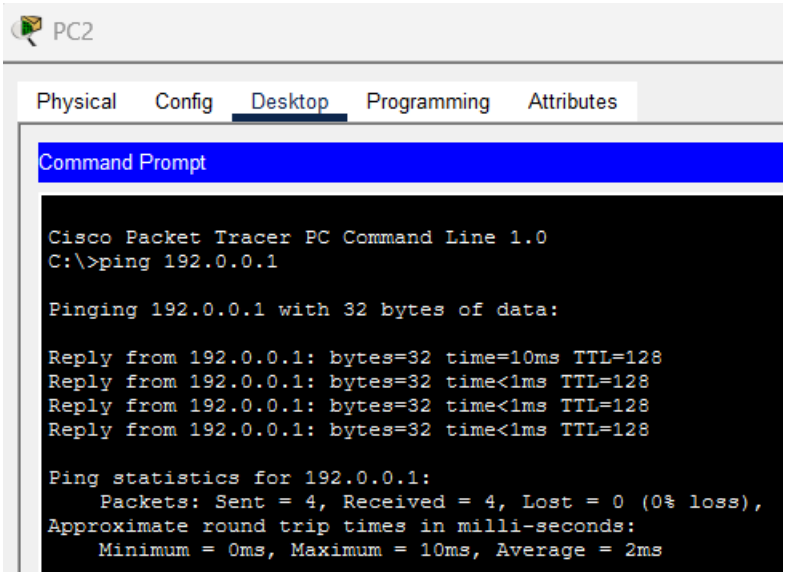


2. MESH TOPOLOGY:

SCENARIO

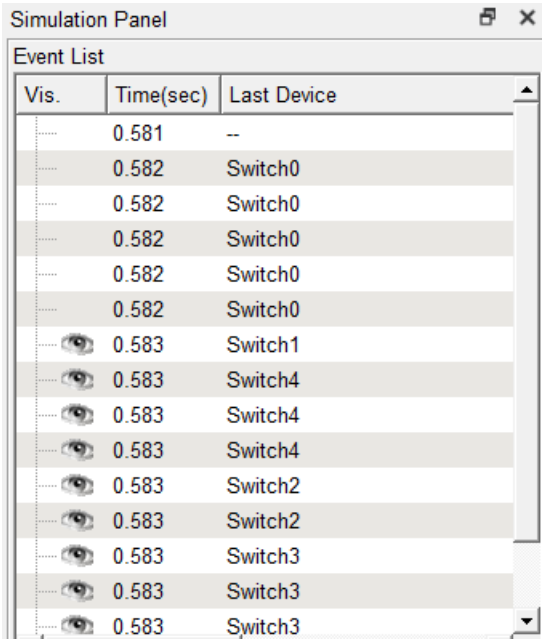


PINGRESPONSE



SIES Graduate School of Technology
RISE WITH EDUCATION

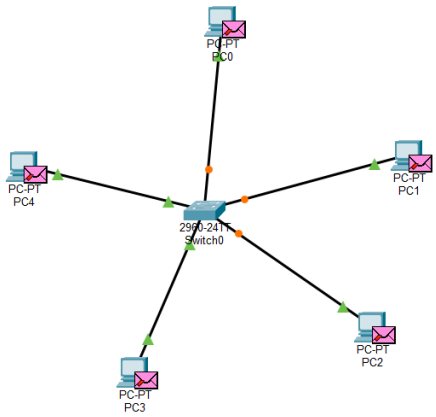
PACKET FLOW SEQUENCE



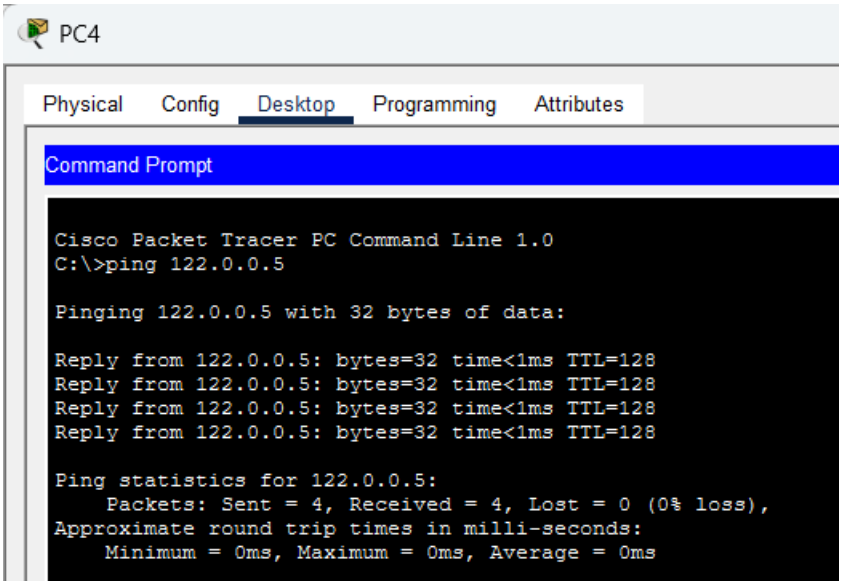
SIES Graduate School of Technology
RISE WITH EDUCATION

3. STAR TOPOLOGY:

SCENARIO

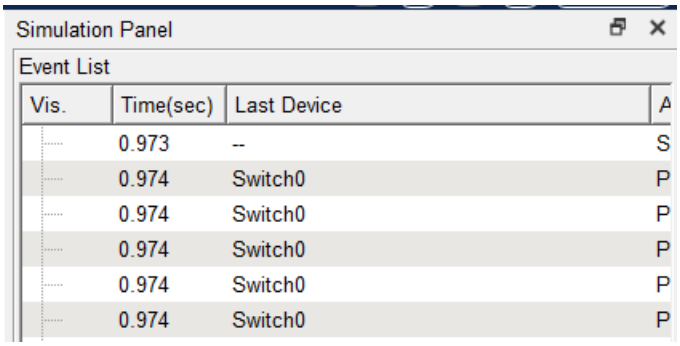


PING RESPONSE:



SIES Graduate School of Technology
RISE WITH EDUCATION

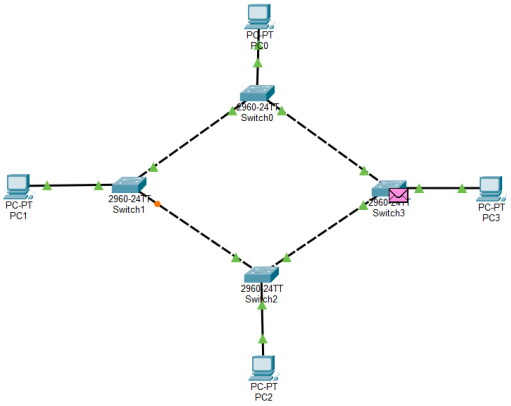
PACKET FLOW SEQUENCE:



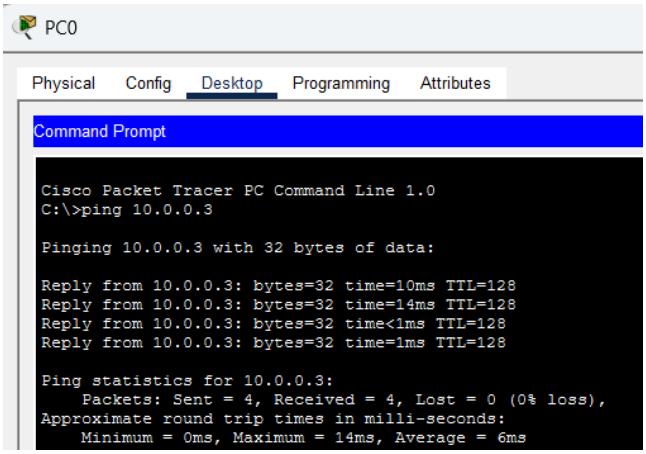
SIES Graduate School of Technology
RISE WITH EDUCATION

4. RING TOPOLOGY:

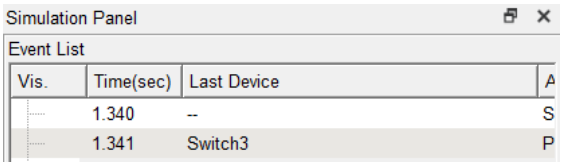
SCENARIO



PING RESPONSE:



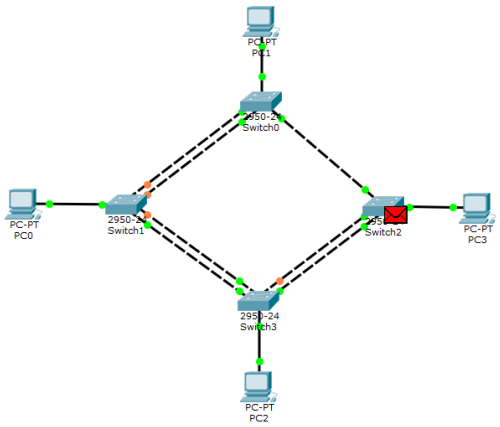
PACKET FLOW SEQUENCE:



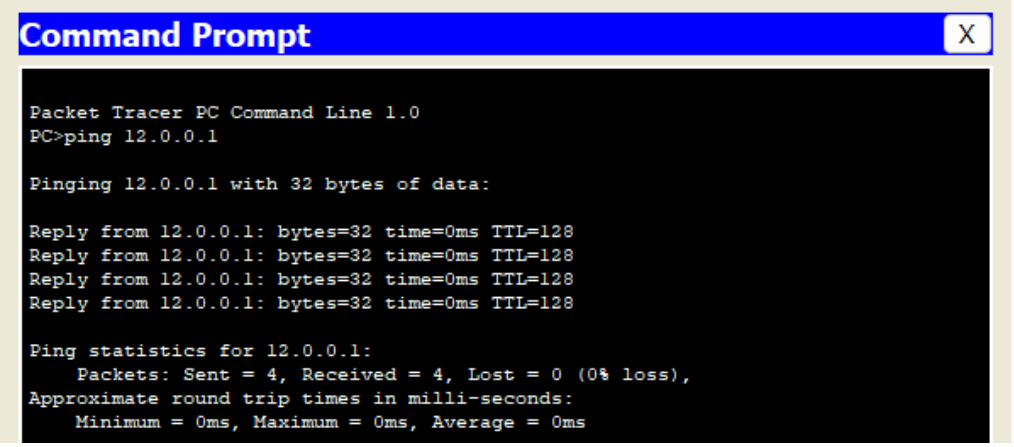
SIES Graduate School of Technology
RISE WITH EDUCATION

5. DUAL RING:

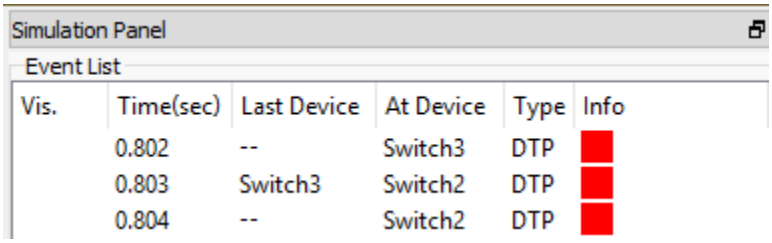
SCENARIO



PING RESPONSE



PACKET FLOW SEQUENCE

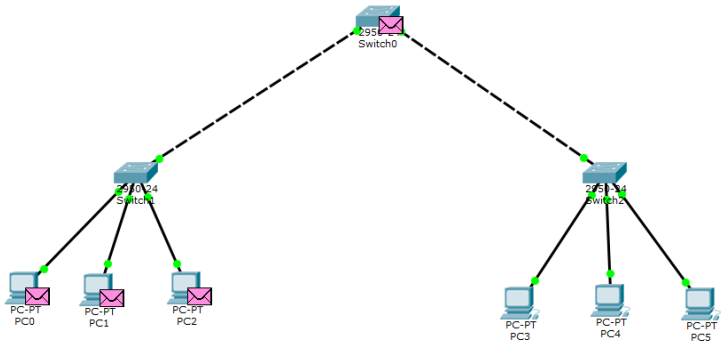



SIES
Graduate School of Technology
RISE WITH EDUCATION

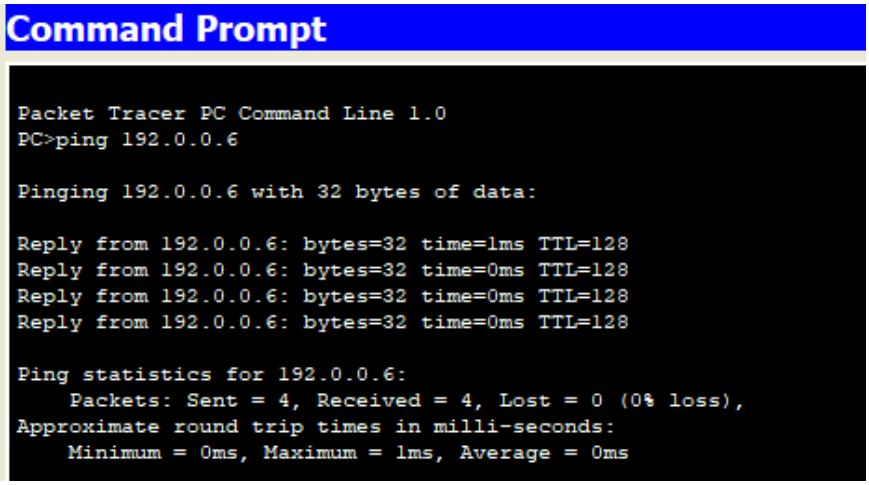
SIES Graduate School of Technology
RISE WITH EDUCATION

6. TREE TOPOLOGY:

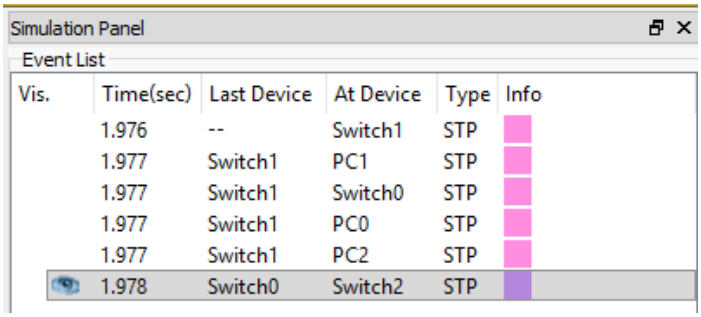
SCENARIO



PING RESPONSE

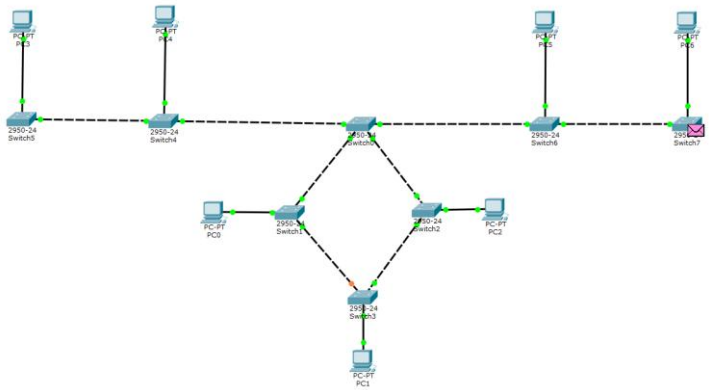


PACKET FLOW SEQUENCE

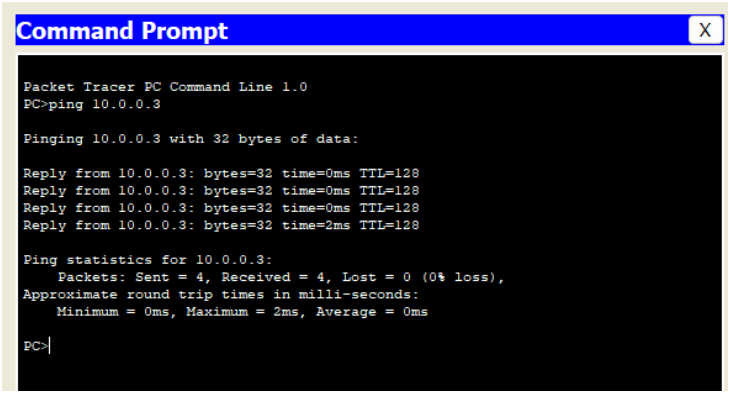


7. HYBRID TOPOLOGY:

SCENARIO



PING RESPONSE



PACKET FLOW SEQUENCE

Vis.	Time(sec)	Last Device	At Device	Type	Info
	1.163	Switch2	Switch3	STP	
	1.163	Switch2	PC2	STP	
	1.163	Switch1	Switch3	STP	
	1.163	Switch1	PC0	STP	
	1.163	Switch4	Switch5	STP	
	1.163	Switch4	PC4	STP	
	1.164	Switch3	Switch1	STP	
	1.164	Switch3	PC1	STP	
	1.164	Switch5	PC3	STP	

SIES Graduate School of Technology
RISE WITH EDUCATION

COMPARISON BETWEEN DUAL RING, TREE and HYBRID TOPOLOGIES.

Parameter	Dual Ring	Tree	Hybrid
Cost	High	Moderate	High
Line Configuration	Two interconnected rings	Hierarchical (parent-child)	Combination of two or more topologies
Installation and Reconnection	Moderate	Easy	Complex
Fault Identification and Isolation	Easy (redundant path available)	Moderate	Easy but depends on topology
Application	FDDI networks, Telecom backbone	Organizational LANs, Campus networks	Large enterprises, Data centers, Industrial networks

SIES Graduate School of Technology
RISE WITH EDUCATION

COMPARISON BETWEEN BUS, MESH, STAR AND RING TOPOLOGIES:

Parameter	Bus	Mesh	Star	Ring
Total Number of Links	*n − 1*	*n(n − 1)/2*	*n*	*n*
Privacy	Low	Very High	High	Moderate
Cost	Low	Very High	Moderate	Moderate
Line Configuration	Single backbone cable	Point-to-point between every node	Dedicated links to central hub/switch	Closed circular loop
Number of I/O Lines with Each Device/Node	1	*n − 1*	1	2
Installation and Reconnection	Easy	Difficult	Easy	Moderate
Fault Identification and Isolation	Difficult	Easy	Easy	Moderate
Application	Small LANs, Temporary networks	Military, Banking, Mission-critical systems	Offices, Schools, Home LANs	Token Ring, Fiber Ring networks

CONCLUSION:-

The experiment demonstrated the working of various network topologies and highlighted their differences in connectivity, reliability, cost, and fault handling. The observations help in selecting an appropriate topology based on network requirements.